

The Operating System for Military Robotics in GPS-Denied Theatres

Hardware-agnostic, zero-external-dependency 3D SLAM, kinematic trajectory planning, and AI threat detection software for Unmanned Ground Vehicles (UGVs).

1,276 / 1,276
TESTS PASSING (100%)

<15ms
HYBRID A* KINEMATIC PLAN

0% Dependency
EMBARGO-PROOF CODEBASE

STANAG 4586
NATO INTEROPERABILITY

THE PROBLEM: EW & ASYMMETRIC LOSSES

- **GPS Denied by EW:** Modern jamming disables satellite navigation within seconds, stranding standard robotics.
- **Lethal Ground Threats:** Mines, IEDs, and tripwires cause 60%+ of UGV losses before reaching mission objectives.
- **Hardware Builders Lack Software:** Vehicle manufacturers build great chassis but lack proprietary autonomy stacks.

THE SOLUTION: FULL-STACK AUTONOMOUS BRAIN

- **GPS-Free 3D SLAM:** Centimeter precision via 3D LiDAR, Visual Odometry, and IMU pose-graph optimization.
- **Kinematic Hybrid A*:** Dynamic route planning respecting real turning radius and slip angles in sub-15ms.
- **AI Threat Fusion:** Real-time GPR/thermal detection creating automatic 30m hazard quarantine buffers.

PRODUCT ARCHITECTURE

1. **Trustia Core:** Perception, SLAM, Hybrid A*, Pure Pursuit.
2. **AI Threat Fusion:** Automated mine/IED/CBRN quarantine rings.
3. **Swarm Intelligence:** UAV-UGV synchronization & formations.
4. **Tactical C2 Console:** MIL-STD-2525 mission dispatch & GUI.
5. **Hardware Bridge:** SAE AS6091 (JAUS), CAN FD, ROS 2 Humble.

BUSINESS MODEL & PRICING

- **Tier 1 (OEM License):** \$15,000 – \$35,000 / vehicle / year.
- **Tier 2 (Swarm C2 Server):** \$50,000 / command center / year.
- **Tier 3 (Defense Integration):** \$250K – \$1.0M NRE engineering.
- **Target Market:** \$34.8B Defense Robotics Market by 2030. Initial SOM: \$450M NATO & Allied Contractors.

COMPETITIVE MATRIX			
Feature	Legacy Defense (Lockheed, GD)	Open Source (Nav2, Autoware)	TRUSTIA AI
GPS-Denied Navigation	Costly (\$500K+ INS Hardware)	Drifts / Unstable in Field	Centimeter 3D SLAM (\$0 GPS)
Mine / IED AI Quarantine	Separate Manual Sensor Kits	None Available	Native 30m Auto Quarantine Ring
Code Rigor & Independence	Black-Box Locked	Fragile Unverified Code	1,276 Passing Verified Tests (100%)

FOUNDER & LEADERSHIP

Murat Furkan Bayram — Founder & Lead Autonomous Systems Engineer
KOSGEB Advanced Entrepreneur (#2026), BTK and Defense Industry Academy certified. Architected the complete sovereign C++20/Python autonomy engine.

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